THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — MATHS · HARD LEVEL UNIT TEST SET 1

Chapters 5–8: **Circles, Perimeter and Area, Probability, Sequences and Progressions** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. Parallel chords of lengths 14 cm and 48 cm lie on opposite sides of the centre of a circle of radius 25 cm. The distance between them (in cm) is: 1

(a) 30 **(b)** 31 **(c)** 32 **(d)** 33

2. The area of a cyclic quadrilateral with sides 5 cm, 7 cm, 15 cm and 19 cm is (cm^2): 1

(a) 95 **(b)** 96 **(c)** 97 **(d)** 98

3. Two dice are rolled together. The probability that the product of the numbers is 12 is: 1

(a) $\frac{2}{9}$ **(b)** $\frac{1}{9}$ **(c)** $\frac{8}{9}$ **(d)** $\frac{1}{10}$

4. The 6th term of an AP is 20 and the 10th term is 32. The 16th term is: 1

(a) 56 **(b)** 53 **(c)** 50 **(d)** 47

5. Parallel chords of lengths 14 cm and 48 cm lie on the same side of the centre of a circle of radius 25 cm. The distance between them (in cm) is: 1

(a) 17 **(b)** 19 **(c)** 18 **(d)** 16

6. A circle of radius 7 cm (take $\pi = \frac{22}{7}$) is inscribed in a square. The area of the square not covered by the circle is (cm^2): 1

(a) 41 **(b)** 43 **(c)** 42 **(d)** 44

7. Two dice are rolled together. The probability that the difference of the numbers is 2 is: 1

(a) $\frac{2}{9}$ **(b)** $\frac{1}{5}$ **(c)** $\frac{7}{9}$ **(d)** $\frac{1}{3}$

8. The sum of the first n terms of a sequence is $S_n = 2n^2 + 3n$. Its 9th term is: 1

(a) 35 **(b)** 39 **(c)** 37 **(d)** 41

9. Two circles of radii 13 cm and 15 cm intersect in a common chord of length 24 cm and their centres lie on opposite sides of it. The distance between the centres (in cm) is: 1

(a) 16 **(b)** 13 **(c)** 15 **(d)** 14

10. Two concentric circles have radii 14 cm and 21 cm (take $\pi = \frac{22}{7}$). The area of the ring between them is (cm^2): 1

(a) 769 **(b)** 771 **(c)** 770 **(d)** 772

11. Three coins are tossed. The probability of getting at least one head is: 1

(a) $\frac{7}{8}$ **(b)** $\frac{1}{8}$ **(c)** $\frac{7}{9}$ **(d)** $\frac{3}{4}$

12. The sum of all integers from 1 to 100 that are divisible by 6 is: 1

(a) 822 **(b)** 810 **(c)** 816 **(d)** 826

13. **Assertion (A):** In a circle of radius 5 cm, a chord at a distance of 3 cm from the centre has length 8 cm. 1

Reason (R): The perpendicular from the centre of a circle to a chord bisects the chord.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

14. **Assertion (A):** The area of a triangle with sides 13 cm, 14 cm and 15 cm is 84 cm^2. 1

Reason (R): By Heron's formula the area is $\sqrt{s(s-a)(s-b)(s-c)}$, where s is the semi-perimeter.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

15. **Assertion (A):** When a fair die is rolled, the probability of getting a prime number is $\frac{1}{2}$. 1

Reason (R): The prime numbers on a die are 2, 3 and 5, which are 3 of the 6 equally likely outcomes.

Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true but R is not the correct explanation of A.

(c) A is true but R is false.

(d) A is false but R is true.

SECTION — B (Very Short Answer Type) 5 × 2 = 10

Answer in about 20–30 words each.

16. Two parallel chords of lengths 14 cm and 48 cm lie on the opposite sides of the centre of a circle of radius 25 cm. Find the distance between them. 2

17. Find the area of a cyclic quadrilateral with sides 5, 7, 15 and 19 cm. 2

18. Two dice are rolled together. Find the probability that the product of the numbers is 12. 2

19. The sum of the first 15 terms of an AP is 255 and the first term is 3. Find d and the 15th term. 2

20. Two circles of radii 10 cm and 17 cm intersect, and the common chord is 16 cm. Find the distance between the centres if they lie on opposite sides of the chord. 2

SECTION — C (Short Answer Type) 5 × 3 = 15

Answer in about 40–60 words each.

21. Two circles of radii 13 cm and 15 cm intersect in a common chord of length 24 cm. Find the distance between their centres if they are on 3

(i) opposite sides,

(ii) the same side of the common chord.

22. A rope 44 cm long is first formed into a circle and then into a square (take $\pi = \frac{22}{7}$). Find the radius of the circle, the side of the square, and which encloses the larger area. 3

23. A bag contains red, blue and green balls. $P(\text{red}) = \frac{1}{3}$ and $P(\text{blue}) = \frac{1}{3}$ and there are 24 balls in all. Find $P(\text{green})$ and the number of balls of each colour. 3

24. The sum of the 3rd and 9th terms of an AP is 40 and its 5th term is 17. Find the AP's first term and the sum of its first 20 terms. 3

OR

Parallel chords of lengths 10 cm and 24 cm in a circle of radius 13 cm. Find the distance between them in both possible positions. 3

25. A running track is made of a rectangle 60 m $\times$ 28 m with a semicircle (radius 14 m) on each of the two shorter sides (take $\pi = \frac{22}{7}$). Find the length of the track along its boundary and the area enclosed. 3

OR

In 100 throws of a die, the number of times a prime number appeared was 52 and a number greater than 4 appeared 31 times (these include 14 throws showing 5). Find the experimental probability of 3

(i) a prime number,

(ii) a number greater than 4,

(iii) a prime number or a number greater than 4.

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

26. **Case study.** A circular garden in a Chandigarh park has radius 13 m. A straight path, 24 m long, joins two points on its boundary (a chord of the circle). 5

(i) Find the half-length of the chord.

(ii) Find the perpendicular distance of the path from the centre.

(iii) Find the length of another chord which is at a distance of 12 m from the centre.

(iv) If the two chords lie on opposite sides of the centre and are parallel, find the distance between them.

27. Two dice are rolled together. Find the probability that 5

(i) the sum is 6,

(ii) the sum is more than 6,

(iii) the sum is at most 6. Verify that

(ii) and

(iii) add to 1.

OR

A contractor pays a penalty for delay: ₹200 on day 1, ₹250 on day 2, ₹300 on day 3 and so on. If the work is 30 days late, find the penalty on the last day and the total penalty. 5

END OF QUESTION PAPER — Total: 50 Marks